

Machine Learning 1 - Practice Problems 6

Universidad de La Sabana
Santiago Toledo-Cortés
Facultad de Ingeniería 2026-2

Instructions

Groups of 3. The assignment must be submitted as a Jupyter notebook. Ensure that the notebook runs and all figures are correctly rendered. The file must be named:

`ml-assign1-username1-username2-username3.ipynb`

For instance, my username is `santiagotoco`, so my file would be named `santiagotoco.ipynb`,

1. Abalone Dataset (1.25 points)

1. Download the Abalone dataset. Ignore the sex attribute and convert the rings attribute into a binary class label `ringClass`, where:

$$\text{ringClass} = \begin{cases} 1, & \text{if rings} \geq \text{average rings} \\ 0, & \text{otherwise} \end{cases}$$

2. Load and process the dataset to obtain NumPy arrays X and y .
3. Split into training and validation using:

`train_test_split(X, y, test_size=0.33, random_state=42)`

4. Train univariate Gaussian classifiers for each feature and evaluate them. Report parameters, accuracy, precision, recall, and draw posterior curves for the best classifier. Also, for the best classifier, plot a graphic showing the decision boundary.

2. Bivariate Gaussian Model (1.25 points)

1. Train bivariate Gaussian models for each pair of features with a shared covariance matrix $\Sigma = I\sigma$, where σ is a scalar. This is, the distribution for each class has a different mean but the same covariance matrix.
2. Estimate parameters and evaluate classifiers in the validation dataset. Which combination of parameters obtained the best performance? Report the evaluation results for the best performing combination.
3. Draw ROC curves for the best univariate and bivariate models. Compare results. Which model is better? Explain the results.

3. Decision Regions (1.25 points)

1. Plot decision regions for the best bivariate model. The regions corresponding to the two classes are shown with different colors. A region corresponding to a class is the set of points where the particular class discriminant function is maximum.
2. The boundary between class regions must be a line. Calculate the equation of this line clearly explaining the deduction process. Draw the line along with the regions.

4. Generalized Bivariate Classifier (1.25 points)

1. Using the best combination of parameters found in question 2 build a bivariate classifier, but this time the covariance matrix could be an arbitrary matrix (not diagonal) and different for each class.
2. Draw the regions corresponding to each class. Compare with the regions obtained in question 3. Discuss.
3. Add the possibility of rejection to your model. Draw the three regions corresponding to both classes and rejection. Discuss.

References

- Alp14 Alpaydin, E. *Introduction to Machine Learning*, 3rd Ed., The MIT Press, 2014.

DHS00 Duda, R. O., Hart, P. E., and Stork, D. G. *Pattern Classification (2nd Edition)*, Wiley-Interscience, 2000.